TABLEAU DATA ANALYSIS PRESENTATION

Accessibility and Completeness of Information about Public Facilities in Tallinn for Wheelchair Users

Tallinn is a city built on convenience. But how accessible is it for wheelchair users?

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ASK: WHAT DID WE WANT TO FIND OUT?

Main research questions:

Question 01

What share of selected LIPS objects in Tallinn are accessible, partially accessible, not accessible, or unknown for wheelchair users?

Question 02

How does accessibility differ between Tallinn districts?

Question 03

Which object categories have better or worse accessibility indicators?

Question 04

How complete is wheelchair accessibility information in OpenStreetMap for the same public facility categories?

PREPARE: WHAT DATA DID WE USE?

Data sources:

LIPS (www.lipseesti.ee)

- Main source for accessibility information
- Contains public facilities and wheelchair accessibility status
- Used as the primary dataset

OpenStreetMap

- Additional open data source (www.openstreetmap.org)
- Used to compare data completeness
- Wheelchair accessibility information is community-based and often incomplete

District Info

Tallinn district information used to analyse accessibility differences between districts

PROCESS: HOW DID WE PREPARE THE DATA?

Main cleaning steps:

1. Downloaded LIPS data
2. Converted JSON data into CSV format
3. Selected relevant columns
4. Grouped detailed LIPS categories into four main categories
5. Standardised accessibility statuses (Accessible, not accessible, partially accessible, unknown)
6. Added Tallinn districts based on coordinates
7. Checked duplicates, missing values, and coordinate quality
8. Prepared summary tables for Tableau visualisation

* Unknown accessibility was kept as a separate category. Unknown does not mean Not accessible.

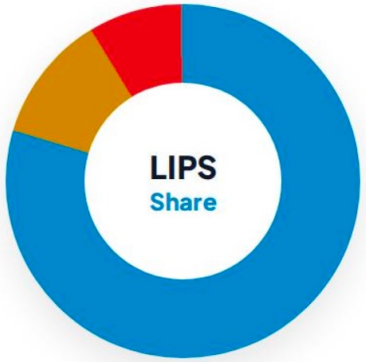
ANALYZE: OVERALL ACCESSIBILITY STATUS IN LIPS

Key finding:

Most selected LIPS objects are marked as accessible.

Accessible	79.63%
Partially accessible	11.69%
Not accessible	8.48%
Unknown	0.19%

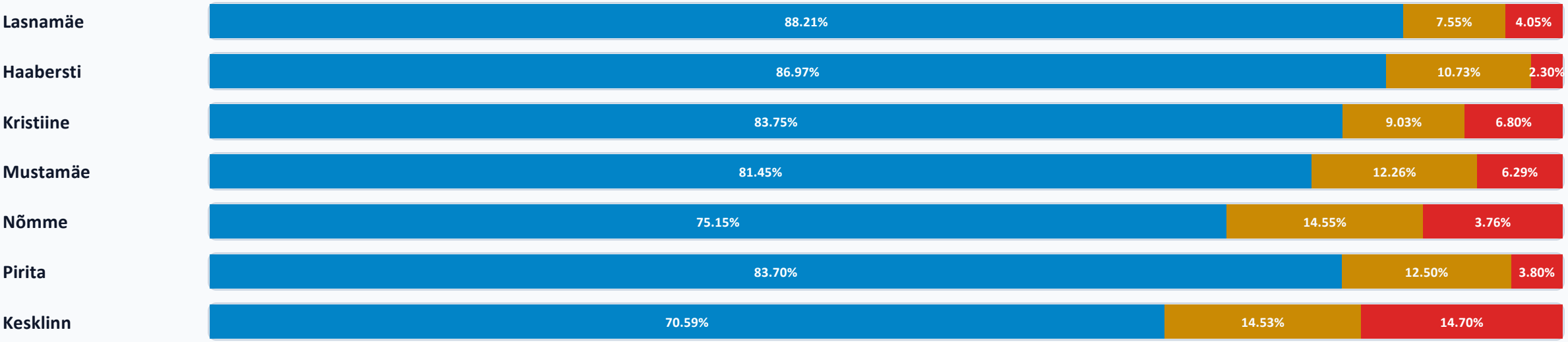
This result describes only selected LIPS objects, not all public facilities in Tallinn.



OVERALL STATUS DISTRIBUTION

ANALYZE: ACCESSIBILITY DIFFERS BETWEEN TALLINN DISTRICTS

Accessibility status of LIPS objects by Tallinn district



- Accessibility is not evenly distributed across Tallinn districts.
- Some districts have a higher share of accessible objects, while others have more partially accessible or not accessible objects.
- This means that accessibility should be analysed not only at the city level, but also by district.

ANALYZE: WHICH DISTRICTS HAVE THE HIGHEST SHARE OF ACCESSIBLE LIPS OBJECTS?

Share of accessible LIPS objects by district



- Lasnamäe, Haabersti and Kristiine have the highest accessible share in the selected LIPS dataset.
- District-level analysis helps identify local accessibility gaps.

ANALYZE: ACCESSIBILITY ALSO DIFFERS BY FACILITY CATEGORY

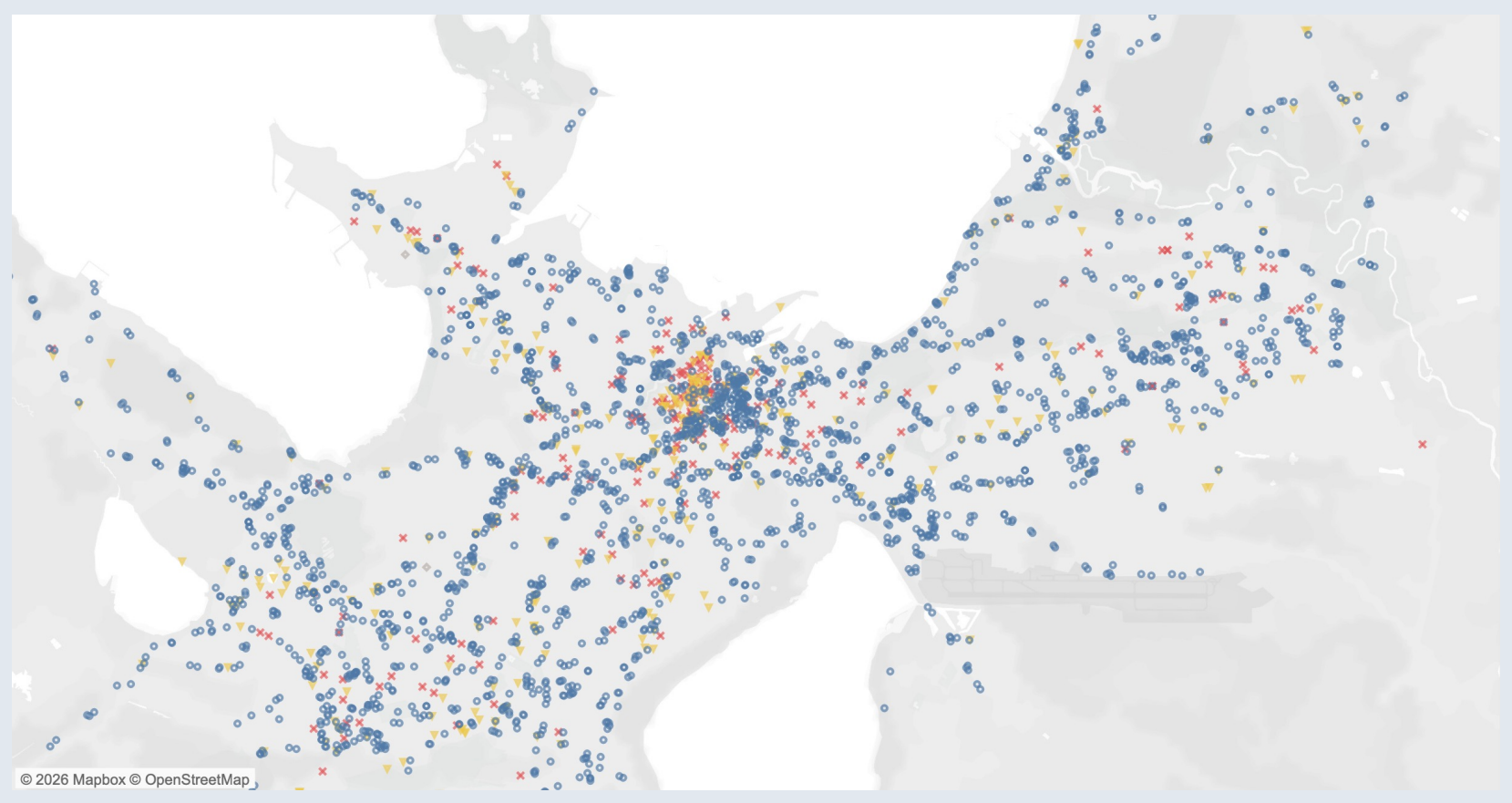
Share of accessible LIPS objects by category



- Daily services have the highest share of accessible objects in the selected LIPS dataset, while Culture / education / leisure has a lower share.
- Accessibility is both a spatial issue and a category-related issue.

SHARE: WHERE ARE ACCESSIBLE AND INACCESSIBLE OBJECTS LOCATED?

Accessibility map of LIPS objects in Tallinn



Category

- Culture/education/leisure
- Daily services
- Healthcare
- Public services

District

- Haabersti linnaosa
- Kesklinna linnaosa
- Kristiine linnaosa
- Lasnamäe linnaosa
- Mustamäe linnaosa
- Nõmme linnaosa
- Pirita linnaosa
- Põhja-Tallinna linnaosa

Accessibility Level

- Full accessibility
- Limited access
- No access
- Unknown

ANALYZE: HOW COMPLETE IS ACCESSIBILITY INFORMATION IN OPENSTREETMAP?

OpenStreetMap completeness check

62.61%

of selected OSM objects have unknown wheelchair accessibility status.

OpenStreetMap contains useful public facility data, but wheelchair accessibility information is often missing.

OSM is useful for comparison, but it is not complete enough to answer accessibility questions on its own.

ACT: CONCLUSION

At the beginning, we asked: "Tallinn is a city built on convenience, but how accessible is it for wheelchair users?"

- **Based on LIPS, many selected public facilities are marked as accessible.**

- **However, the available data is not sufficient to conclude that Tallinn as a whole is fully accessible for wheelchair users.**

- **The project shows that accessibility data itself is part of accessibility. If information is missing, unclear, or fragmented, wheelchair users still face barriers.**

ACT: RECOMMENDATIONS

Recommendations

1. Keep LIPS accessibility data updated
2. Add more detailed accessibility features where possible
3. Improve wheelchair accessibility tags in OpenStreetMap
4. Use district-level analysis to identify local accessibility gaps
5. Help city services prioritise accessibility data improvements

Thank you!



Thank You!

Questions?